

Koopman State Residual Approximation for Model Predictive Vehicle Trajectory Tracking

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Abstract—Abstract—This study employs Extended Dynamic Mode Decomposition (EDMD) to learn residual vehicle dynamics, addressing the challenges posed by inaccurate nominal model parameters and unmodeled physics. By identifying residual state dynamics via Koopman operators in a lifted space and projecting them back onto the original state, we augmented the nominal model to enhance the overall predictive accuracy. A Model Predictive Controller (MPC), implemented using CasADi, utilizes this augmented model for trajectory tracking. Experimental results demonstrate that the proposed framework achieves a tracking error of less than $0.6m$, even in the presence of a significant model mismatch. To ensure ride comfort during aggressive maneuvers, we incorporated state- and input-smoothing penalties to reduce the jerk. The controller generates optimal control signals that maintain the peak jerk below $1.5m/s^3$ initially and below $0.5m/s^3$ during steady-state sinusoidal trajectory tracking. In future work, this approach will be compared with the Residual Koopman MPC and its performance will be evaluated in high-fidelity simulation environments.

I. INTRODUCTION

Accurate trajectory tracking is a fundamental requirement for autonomous driving in both on-road and off-road conditions. Conventional trajectory-tracking approaches, such as Pure Pursuit [1], often employ model-free algorithms, making them simple, computationally efficient, and attractive for real-time implementation [2]. These methods rely on a single time-step preview and cannot optimize control inputs for global trajectory tracking. Moreover, because they operate over short prediction intervals and neglect vehicle model constraints, the resulting control commands may exceed the physical limits, making it difficult to follow the desired trajectory and potentially compromising safety [3]. Their performance also deteriorates under conditions such as heavy load variations, high-speed operation, and complex tire–terrain interactions.

In contrast, model-based predictive approaches leverage a vehicle model to forecast future states and optimize control inputs while explicitly accounting for constraints over the prediction horizon [4]. Consequently, effective trajectory tracking in complex dynamic environments, particularly at high speeds, requires an accurate representation of vehicle dynamics to ensure safe and stable maneuvers.

Model Predictive Control (MPC) implementations often rely on linearized or simplified vehicle dynamics. Although linearization produces computationally efficient models, it introduces inaccuracies [2], limiting the performance of linear MPC, which is highly dependent on model fidelity. Linearized models frequently fail to capture the nonlinear properties farther from the equilibrium points that emerge

under extreme operating conditions, such as high lateral acceleration and low friction surfaces. Furthermore, vehicle parameters, including tire stiffness, friction coefficients, and load distribution, vary with operating conditions, which can further degrade MPC performance if not updated appropriately.

Nonlinear Model Predictive Control (NMPC) overcomes the limitations of linear MPC by explicitly incorporating nonlinear vehicle dynamics [5]. However, the time-varying nature of vehicle dynamics requires real-time updates to the model. NMPC is computationally demanding and often exceeds the capabilities of onboard computing. Since both NMPC and MPC require an accurate model, they need a means to update their model parameters to accurately describe the vehicle.

Recent studies have investigated the transformation of NMPCs into Koopman-based MPC frameworks by linearizing the vehicle state space using Koopman observables [6]. In this approach, nonlinear vehicle states are lifted into a higher-dimensional space using a set of basis functions. This formulation mitigates the trade-off inherent in conventional linearization methods, which often compromise the model accuracy to achieve computational efficiency. Furthermore, Koopman-based modeling has been combined with stochastic MPC for lane-tracking applications [7], where the Extended Dynamic Mode Decomposition (EDMD) approximates the Koopman operator using system observables. By contrast, the stochastic MPC explicitly accounts for modeling uncertainties. These approaches assume known vehicle dynamic model parameters and lack adaptation to changes in vehicle properties.

In this study, we integrated the nominal model of the vehicle and employed the Koopman operator to learn the residual state dynamics. The enhanced vehicle model consisted of a nominal model with fixed parameters and a component linearized by the Koopman operator, which was initially set to zero. The residual state dynamics are continuously updated through a lifted high-dimensional representation in the Koopman operator, which captures and adapts to the parameter variations and compensates for the nominal model inaccuracies. The Koopman-based model is constructed using recursive least-squares estimation, allowing for more efficient real-time updates of the Koopman model. [8]. As the Koopman model is progressively learned from the measurement data, it improves the representation of the system dynamics and accuracy. Consequently, our approach enables the MPC to improve tracking performance under significant parameter variations.

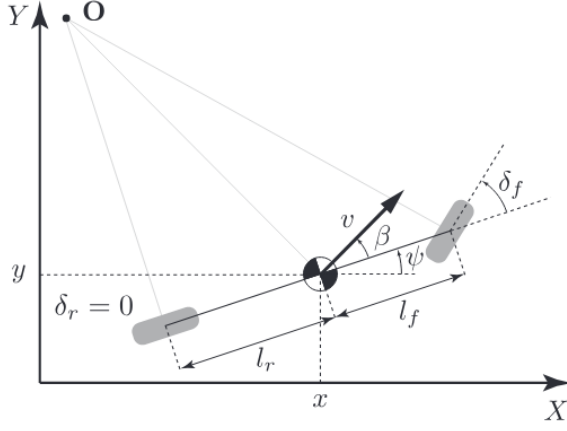


Fig. 1. Bicycle model of vehicle

The remainder of this report is organized as follows. Section (II) discusses the problem formulation. Section (III) discusses the experimental settings. Section (IV) discusses the performance evaluation of the Koopman residual state dynamics in the vehicle trajectory tracking problem. Section-V presents the summary and discussion.

II. METHODS

A. System Modeling

In this study, the nominal vehicle system dynamics were modeled using a bicycle model, which was chosen for its simplicity. This nominal model operates under the assumption of fixed, known parameters, with any deviations from these parameters addressed by the Koopman residual-state dynamic model. Figure 1 [2] illustrates the bicycle model of the vehicle. The nominal vehicle model comprises six state variables and two input signals, as shown in Equation 1. Although our model assumes a flat surface, it can be adapted to incorporate motion along the Z axis.

$$\frac{d}{dt} \begin{bmatrix} X \\ Y \\ \psi \\ v_x \\ v_y \\ r \end{bmatrix} = \begin{bmatrix} v_x \cos(\psi) - v_y \sin(\psi) \\ v_x \sin(\psi) + v_y \cos(\psi) \\ r \\ r v_y + a_x \\ -r v_x + 2(F_{cf} \cos(\delta_f) + F_{cr})/m \\ 2(l_f F_{cf} - l_r F_{cr})/I_z \end{bmatrix} \quad (1)$$

Here X , and Y are the positions in the global reference frame. v_x and v_y are the vehicle's longitudinal and lateral speeds, respectively, in the body frame. ψ is the yaw angle, r is the yaw rate of the vehicle, m is the mass of the vehicle, I_z is the inertia of the vehicle. The rear and front wheel cornering stiffnesses are denoted by C_{α_r} and C_{α_f} , generating lateral forces given by $F_{cr} = C_{\alpha_r} \alpha_r$ and $F_{cf} = C_{\alpha_f} \alpha_f$ for the slip angle of α_r/f .

The parameters used in the nominal model of the vehicle dynamics in Equation 1 significantly vary with the operating conditions. For instance, the mass of a vehicle changes with the load, the center of mass may shift depending on the

load placement, and the cornering stiffness varies with the tire type and change in tire pressure. To account for model mismatch, we employed the Koopman operator to learn the residual state dynamics from the measurement data.

B. Koopman Residual State Dynamics

We consider the nonlinear discrete-time representation of a system $x_{k+1} = f(x_k, u_k)$ where x refers to state variables $[X, Y, \psi, v_x, v_y, r]^T$ and steering angle and acceleration are inputs, $u = [\delta, a_x]^T$. Given the observation of the function of $g(x_k)$, the system state is lifted from a finite $\mathbb{R}^N$ dimensional space to a higher-dimensional space, $\mathbb{R}^M$, with the Koopman operator of K as follows.

$$Kg(x_k) = g(f(x_k, u_k)) \quad (2)$$

The high-dimensional lifted space model in Equation 2 represents an infinite-dimensional space. The finite-dimensional approximation of the Koopman operator is obtained using the least-squares estimation algorithm, as in [3]. Continuous observation of the state and input is used to find the lifted high-dimensional space z_k , which is then used to compute an approximate Koopman operator. The finite-dimensional lifted state-advancing matrices $\hat{A}$ and input matrix $\hat{B}$ are obtained by minimizing the state-advancing error and are continuously updated online.

$$(\hat{A}, \hat{B}) = \underset{A, B}{\operatorname{argmin}} \sum_{k=1}^K \|z_{k+1} - (Az_k + Bu_k)\|^2 \quad (3)$$

where $z_k = [x_k^T, v_x^2, v_y^2, r^2, v_x v_y, \sin \psi, \cos \psi]^T$. The lifted. The residual state, x_{rk} , in the original space, can be recovered from the high-dimensional state by using the relation established in Equation 4

$$\hat{C} = \underset{C}{\operatorname{argmin}} \sum_{k=1}^K \|(x_{k+1}^{obs} - (A_k x_k + B_k u_k)) - C z_{k+1}\|_I^2 \quad (4)$$

To this end, we assume that all state variables in the original space are observable from the measurements. The residual state dynamics at any sample time k can be computed in terms of a high-dimensional lifted space state z_k .

$$z_{k+1} = \hat{A}_k z_k + \hat{B}_k u_k \quad (5)$$

$$x_{r, k+1} = \hat{C} z_{k+1} \quad (6)$$

Figure 2 shows a schematic of the application of Koopman's higher-dimensional approximation to learn the residual state dynamics of the vehicle. The complete system dynamics in the original state, augmented with the residual-state dynamics from Equation 6, can be combined to yield Equation 7. The Koopman residual-state dynamic model compensates for the model mismatch and other unmodeled dynamics that the nominal vehicle model does not capture. Therefore, the augmented model best approximates the true dynamics of

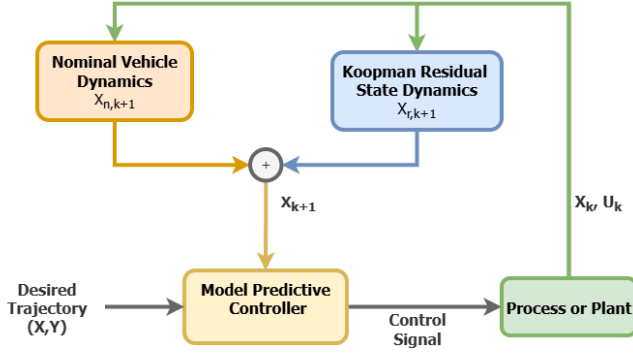


Fig. 2. Schematic of the Koopman-MPC framework for trajectory tracking

the vehicle and enables the model predictive controller to generate an optimal control signal, $u_k = [\delta, a_x]$, to drive the vehicle along the desired trajectory.

$$\begin{aligned} z_{k+1} &= \hat{A} \cdot z_k + \hat{B} \cdot U_k \\ x_{k+1} &= A \cdot x_k + B \cdot u_k + \hat{C} z_{k+1} \end{aligned} \quad (7)$$

C. Model Predictive Control

A model predictive controller is designed to minimize the deviation between the predicted and reference trajectories while accounting for constraints or bounds on the predicted state. In addition, it consists of state and control input smoothing penalties. The smoothing cost aims to mitigate jerk and improve passenger comfort. The MPC controller generates the optimal steering angle and acceleration/throttle that satisfy the constraints on both the states and inputs over a given time horizon. The objective function of a model predictive controller uses augmented model of the vehicle leveraging Koopman residual state dynamics to get high accurate prediction 8

$$\begin{aligned} u_k^* &= \underset{u_k}{\operatorname{argmin}} \sum_{k=0}^{N-1} \|x_k - x_k^{ref}\|_Q^2 + \|u_k\|_R^2 \\ &\quad + \|x_k - x_{k-1}\|_{Q_\Delta}^2 + \|u_k - u_{k-1}\|_{R_\Delta}^2 \\ \text{s.t. } &x_k \in [x_{min}, x_{max}], u_k \in [u_{min}, u_{max}] \end{aligned} \quad (8)$$

where x_k^{ref} is the desired trajectory and u_k is the control effort. Q, R, Q_Δ and R_Δ are the weight matrices related to trajectory tracking error, control effort, and state and input smoothing cost. $[u_{min}, u_{max}]$ refers to the upper and lower bounds of the input signal, respectively, and $[x_{min}, x_{max}]$ refers to the upper and lower limits of the state vectors in the original space, respectively.

Nonlinear vehicle dynamics were used to generate a sufficiently excited dataset for training the Koopman residual state dynamics model. First, the original states are lifted to a higher-dimensional space, and the state advancement and input matrices for the higher-dimensional representation are learned. Then, the residual state $x_{r,k}$ is obtained, and the matrix $\hat{C}$ that maps the higher-order functional space z_k into the residual state of the original space is identified.

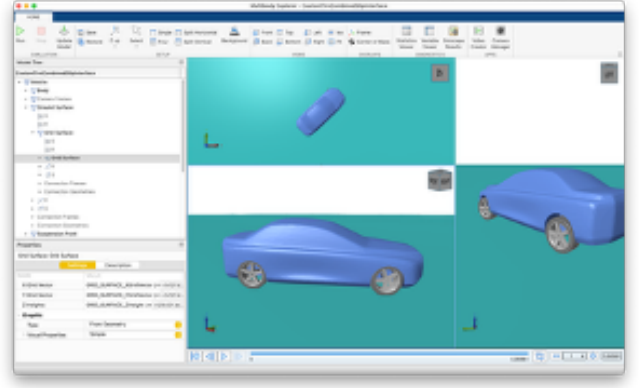


Fig. 3. High fidelity simulation model

With this approach, the model mismatch between the true vehicle and nominal model is approximated using Koopman operators without the need for explicit system identification. The augmented model is then used in a model predictive controller that computes an optimal control input to track the desired state trajectories.

III. EXPERIMENT SETTING

The experiment was designed on a simulation platform to evaluate the proposed approach. The simulation and model considered a constant rate sampling, $\Delta t = 0.05 \text{ sec}$ and the nominal vehicle parameter set is shown in Table I. The true dynamics of the vehicle use different parameters, which will be discussed in the subsequent section.

TABLE I
NOMINAL VEHICLE PARAMETERS

Parameter	Value
Weight(m)	1500kg
Inertial (I_z)	3000kg.m ²
Cornering stiffness ($C_{\alpha f}, C_{\alpha r}$)	19kN/rad, 33kN/rad
Wheel base	2.8m
Centroid (l_f, l_r)	1.2m, 1.6m

The model predictive controller employs the following state-tracking error, control effort, and state and input smoothing penalty weights: The state tracking error weight Q was set to $\text{diag}([5.0, 8.5, 0.0, 0.0, 0.75, 0])$, and R was set to $\text{diag}([1.0, 1e-3])$. The state and input smoothing-related penalties are $Q_\Delta = \text{diag}([0.0, 0.0, 0.0, 0.5, 1.85, 0.05])$ and $R_\Delta = \text{diag}([10.5, 0.5])$.

IV. RESULTS AND DISCUSSION

We designed two experiments to evaluate the performance of the Koopman residual state dynamics model and the designed model predictive controller. The first experiment considers a moderate deviation between the nominal vehicle model and true vehicle dynamics, whereas the second case assumes a greater parameter mismatch in the nominal vehicle dynamics model.

Case 1: In this experiment, the vehicle mass and inertia were varied by 30% and 16% respectively, relative to their nominal

values. The model predictive controller was tasked with following a sinusoidal trajectory that required continuous steering input. The control effort generated by the MPC was then used to drive the vehicle, and its trajectory tracking performance was evaluated.

Figure 4 shows the trajectory-tracking performance of the MPC controller using an augmented model. As shown in the figure, the Koopman residual state-dynamics model can learn and compensate for the nominal model error. Consequently, the MPC obtains a more accurate predictive model over the given prediction horizon and can drive the vehicle along the desired trajectory.

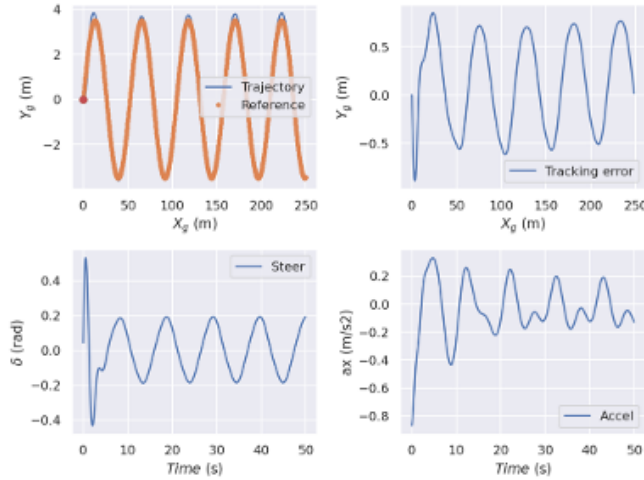


Fig. 4. Trajectory tracking performance and control effort (Case-1)

As shown in Figure 4 the proposed method was able to achieve a trajectory tracking error of less than $0.6m$ while maintaining the steering and acceleration within the bound. We evaluated the jerk and lateral acceleration resulting from the control effort to ensure passenger comfort and compliance of the model with standards, as shown in Figure 5. The proposed method generates a longitudinal jerk that is less than $2m/s^3$ even at the beginning of the trajectory. This is a result of the need to accelerate to reach the optimal speed. Once the vehicle reaches the speed and starts tracking the trajectory, the jerk falls below $0.4m/s^3$ despite a need for continuous maneuvers and throttle effort to follow sinusoidal trajectory.

We expect improved model performance from incorporating online learning; however, edge cases may not have sufficient data to yield meaningful improvements. This can lead to overfitting of a limited set of scenarios in the observations. Therefore, appropriate weighting or regularization strategies may be necessary to ensure a robust generalization.

Case 2: In this experiment, additional parameter variations were introduced to further evaluate the performance of the proposed approach. The cornering stiffness values of the vehicle were varied by 50% relative to the nominal model, in addition to 33% and 16% deviations in the vehicle mass and inertia, respectively. Figure ?? shows the resulting trajectory-tracking performance of the vehicle and the optimal control

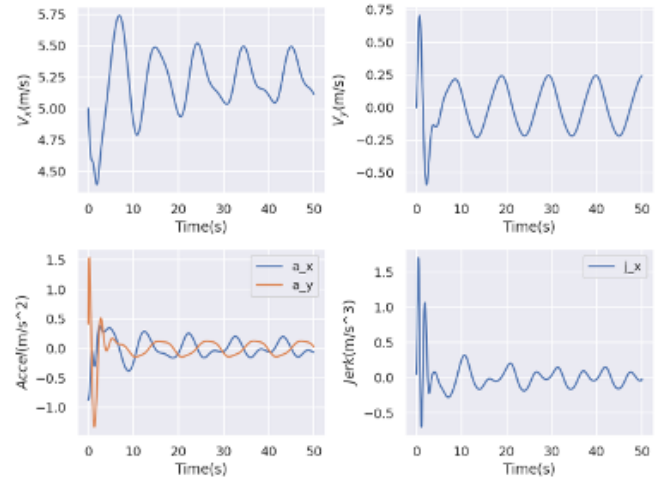


Fig. 5. Acceleration and jerk of the vehicle (Case-1)

effort generated by the MPC controller that uses the Koopman residual-augmented model to predict the future vehicle states. As shown in Figure ??, despite the significant deviation from the nominal model, the Koopman residual state-dynamics model can compensate for the model mismatch and provide the MPC controller with a more accurate predictive model. Consequently, a peak trajectory-tracking error of less than $0.6m$ was achieved.

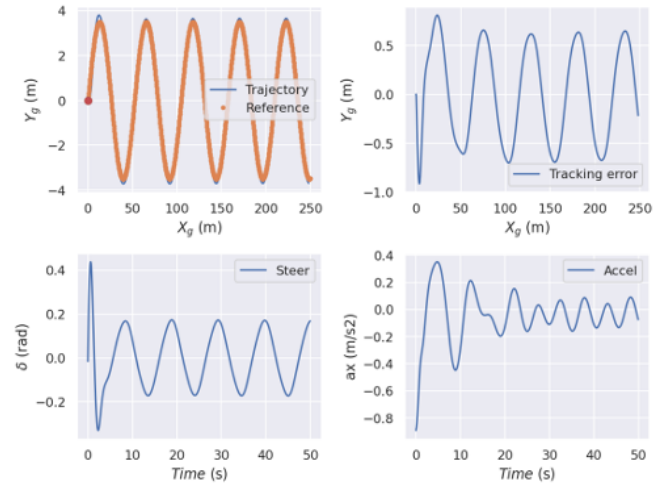


Fig. 6. Trajectory tracking performance and control effort (Case-2)

Figure 7 shows the details of the trajectory tracking performance relative to passenger comfort. The proposed method generates an optimal control effort that maintains the longitudinal jerk below $2m/s^3$ and drives the vehicle with a low lateral acceleration. At the beginning of the trajectory, the model predictive controller generated an aggressive control effort to drive the vehicle along the desired trajectory and achieved a velocity suitable for following the reference trajectory. This causes an increased jerk close to $1.55m/s^2$; however, as the vehicle starts following the trajectory, it quickly drops below $0.5m/s^3$.

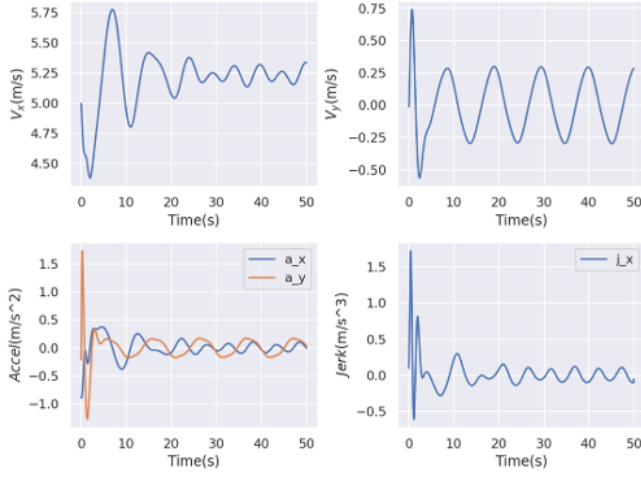


Fig. 7. Acceleration and jerk of the vehicle (Case-2)

V. SUMMARY AND CONCLUSION

In this study, a nominal nonlinear state-space model was used to generate the training data. An initial nonlinear model was identified from the collected data and used to control a test model whose parameters deviated from their nominal values. The resulting prediction errors from the test model were used to capture the residual nonlinear dynamics, which were then fitted into a Koopman-based residual model. The performance of the augmented model, which consists of the nonlinear model and the Koopman residual correction, is presented in the results.

Future work will focus on further evaluation and extension of the proposed approach. First, the proposed Koopman-based residual modeling framework can be integrated with a Koopman MPC controller, which enables predictive control in a linear-lifted state representation. The resulting performance can then be compared with that of the current nonlinear MPC framework to evaluate the potential advantages in prediction accuracy, computational efficiency, and closed-loop control performance. This comparison helps determine whether representing the system dynamics in a lifted Koopman space provides practical benefits for controlling the augmented residual model.

Second, the proposed method can be tested using a high-fidelity model to better evaluate its performance under realistic operating conditions. High-fidelity simulation environments, such as AutoDRIVE and CARLA, would allow the residual model to capture more complex effects, including unmodeled nonlinearities, parameter variations, disturbances, and actuator limitations.

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